

Smart Heart Risk Analyzer with Blockchain Verification

1 Monika B N, 2 Spoorthi K, 3 Varsha M R, 4 Lavanya B C Department of Artificial Intelligence & Data Science, S J C Institute of Technology, Chikkaballapur, India, Email: gowdemonikan147@gmail.com, spurthikaradakatti@gmail.com, varshagowda010@gmail.com, bclavanya2003@gmail.com.

Abstract: Heart disease is a leading cause of death in people worldwide, making early diagnosis and treatment essential. This study proposes a secure web application for heart disease risk prediction based on blockchain technology. The web application is built with the Flask framework. It consists of a machine learning model that was trained on clinical and physiological data and a blockchain based user registration system. Users can authenticate on the platform and register their Ethereum based identity on-chain. Using their self-sovereign identity, they can input their health data including age, heart rate variability, blood pressure and lifestyle factors. The inputs are passed through the predictive model, either a trained machine learning model or a Keras based deep learning model. After prediction, the model's performance metrics are displayed. The algorithm implements predictive analytics and a blockchain-enabled security and user management system allowing the app to deliver an unbiased, tamper-proof and secure assessment of a person's cardiovascular health.

Key Words: Blockchain, Fuzzy Logic, Machine Learning, Disease Prediction.

1. INTRODUCTION: Cardiovascular diseases (CVDs) are among the most prevalent and life-threatening health conditions worldwide, accounting for a significant proportion of global mortality. As heart disease often benefits from early treatment, methods for detecting the disease are available. Conventional diagnoses of heart disease are based on clinical examination and laboratory investigations, and either must be interpreted from the patient's history and medical imaging. These methods are time consuming and can be inaccurate.

With the advancement of artificial intelligence and machine learning, prediction algorithms can be built to analyze big data in the health sector to identify individuals at high risk for heart disease, though privacy, security, and data integrity issues are raised in collecting data for these purposes. Blockchain could address these issues, the integration of blockchain technology ensures a decentralized, tamper-proof, and transparent system for storing user credentials and critical health records.

This presents a web-based heart disease detection system built on the Flask framework, which combines machine learning-driven prediction with blockchain-enabled secure registration. Users can securely register, submit health-related parameters, and receive personalized risk assessments, while their registration data and transaction records are stored immutably in the blockchain. This integration of AI and blockchain enhances the reliability of predictions and strengthens trust in the management of sensitive health information, offering a robust platform for proactive cardiovascular care.

2. LITERATURE REVIEW: The literature study confirmed that blockchain technology is a valid approach for protecting patient data involving IoMT networks, as it is a secure and decentralized approach against unauthorized access, as well as a transparent and immutable system. On the other hand, fuzzy logic has handled, has interpreted, and has classified unpredictable, uncertain, vague, incomplete, or fuzzy information within healthcare, enabling accurate prediction of diseases. Both technologies promise, but someone has not studied how they combine extensively.

While a few proposals have addressed security and prediction together, most literature focuses on only one of these domains in particular, or doesn't fully address both real-time and IoMT support. Other challenges include limited efficiency, scalability, and lack of flexibility for integrating a wide variety of IoMT devices into a predictive security system. Overcoming these challenges requires a hybrid architecture incorporating secure data management and smart reasoning capabilities.

The survey results conclude that a blockchain and fuzzy logic-based secure disease prediction model can overcome secure and accurate predictive analytics problems of IoMT data in the healthcare application, and can provide reliable and clever healthcare solutions at scale while ensuring the patient security and data integrity with timely medical information delivery. It justifies the need for the proposed project and directs its design and implementation.

3. OBJECTIVES / AIMS: The primary objectives of this study are as follows:

- Implement blockchain-based registration to ensure the protection of sensitive patient information.
- Track and display model performance metrics such as accuracy, precision, and recall to ensure reliability.
- It provides users with a web interface to input health parameters and receive immediate, personalized heart disease risk predictions.
- Combining machine learning for predictive analytics with blockchain technology create a secure and trustworthy platform.

4. RESEARCH METHOD / METHODOLOGY: We built the model using a step-by-step process. We train the model through data scraping, data cleaning and ML modules training.

4.1 Data Acquisition and Preprocessing Module: Data acquisition module (DAQ) is a crucial subsystem in any data measurement, control, or monitoring systems due to its role as the transition between physical world and digital processing system. The primary role is to acquire real life data from the different physical processes such as temperature, pressure, light intensity, sound, force or motion using sensors or transducers and then process these data into a digital form which can be further processed for analysis, storage or transmission.

4.2 Disease Prediction Module: The Disease Prediction Module is a smart computing machine used to process different medical data and predict the onset, progression, or relapse of the disease. It is an important decision-support system in contemporary health care, because the early detection of high-risk status of a patient can materialize into effective intervention that helps to reduce the risk. By combining machine learning methods, statistical modelling and domain specific clinical insight the module is able to find subtle patterns in patient data that are often difficult for a human clinician to manually identify.

4.3 Blockchain Security Module: The Blockchain-Based Security Module is an advanced security architecture designed to safeguard digital information by harnessing the core principles of blockchain technology. Its primary objective is to provide strong data integrity, confidentiality, and accountability across a distributed environment in which multiple participants interact and exchange data. In traditional systems, security and verification depend on a centralized authority, such as a server or administrator, which is responsible for validating user actions, authorizing access, and maintaining database consistency.

5. FINDINGS / RESULTS:

5.1 Login and Dashboard Output: Secure login validation with session initiation. Access to a personalized dashboard displaying user details, prediction options, and performance metrics.

5.2 Prediction Output: Probability score in real-time that shows how likely user is to have heart disease based on his/her health parameters. Suggesting a list of risk labels such as High Risk, Low Risk. The input values, and the predicted outputs are shown for clarity.

5.3 Model Metrics Output: Model performance metrics such as accuracy, precision, recall, F1-score and ROC curves. Multiple model comparison (when applicable) for an indication of prediction reliability.

6. DISCUSSION / ANALYSIS: Performance analysis assesses the efficiency, reliability, and scalability of the system under different conditions. Metrics such as response time, data throughput, prediction latency, and blockchain transaction time were measured during continuous operation with multiple IoMT devices. The results show that the system can process real-time data from multiple patients without significant delays, ensuring timely disease prediction and alert generation.

The security performance was also evaluated by simulating unauthorized access attempts, data tampering, and smart contract violations. The blockchain layer successfully prevents unauthorized actions and maintains data integrity and confidentiality. Additionally, fuzzy logic predictions maintain high accuracy even with noisy or incomplete datasets. This demonstrates the robustness and reliability of the integrated approach, which combines IoMT monitoring, blockchain security, and intelligent inference.

7. CONCLUSION / SUMMARY: The Heart Disease Detection system presented in this project successfully integrates machine learning and blockchain technology to provide a secure, reliable, and user-friendly platform for early cardiovascular risk assessment. With predictive algorithm and registration mechanism based on blockchain, the validity of prediction results remains unaffected but also the security of private data is

guaranteed. Users can sign up with ease, take health metrics and get in-the-moment risk assessments on their health data to hosting tamper-proof and traceable copies of their medical information on a blockchain.

8. ACKNOWLEDGMENT: The computer facilities required for this work were provided by the Department of Artificial Intelligence & Data Science. The authors are grateful. The authors also acknowledge the authors who have contributed the open-source community.

REFERENCES:

1. K. Sharma, S. Chen, and J. Park, "Blockchain-Based Secure Data Sharing for IoMT Applications," *IEEE Access*, vol. 8, pp. 212345–212356, 2020.
2. A. Peyvandi, B. Majidi, S. Peyvandi, and J. Patra, "Computer-aided- diagnosis as a service on decentralized medical cloud for efficient and rapid emergency response intelligence," *New Gener. Comput.*, vol. 39, no. 3–4, pp. 677–700, 2021.
3. S. Agarwal and M. Jain, "IoMT-Enabled Healthcare Systems: A Review," *Journal of Medical Systems*, vol. 44, no. 12, pp. 215, 2020.
4. K. Datta Gupta, D. K. Sharma, S. Ahmed, H. Gupta, D. Gupta, and C.-H. Hsu, "A novel lightweight deep learning-based histopathological image classification model for IoMT," *Neural Process. Lett.*, vol. 55, no. 1, pp. 205–228, 2023.
5. N. Patel, V. Sharma, and A. Kumar, "Real-Time Disease Prediction using IoT and Fuzzy Logic," *Procedia Computer Science*, vol. 167, pp. 1281–1290, 2020.
6. H. Kim and J. Park, "Smart Healthcare with IoMT: Security and Privacy Challenges," *Future Generation Computer Systems*, vol. 108, pp. 1036–1047, 2020.
7. R. Tripathi and S. Singh, "Fuzzy Logic-Based Intelligent Decision Support System in Healthcare," *International Journal of Fuzzy Systems*, vol. 22, no. 4, pp. 1231–1242, 2020.